

SUMMER SCHOOL ON DATA-DRIVEN MODELLING FOR MECHANICAL SYSTEMS



Mr. Chandra Bhushan
Chief Iron Making Mechanical maintenance TSJ,
Chief Shared Services, TSMD Kharagpur, India

With nearly 27 years of experience at Tata Steel, Mr. Bhushan specializes in asset management, reliability engineering, and operational excellence. An alumnus of BIT Mesra, IIT Kharagpur, and XLRI, he leads major initiatives in life-cycle optimization and sustainable manufacturing. During his session, he will explore the evolution of industrial robotics, presenting real-world case studies on automation, digital transformation, and smart factories in large-scale industrial environments.



Dr. Sundar Namala
Application Engineer
COMSOL, India

Dr. Sundar, an Application Engineer at COMSOL with a Ph.D. from UIUC, specializes in computational fluid dynamics for complex multiphysics systems. His session will explore how a unified simulation environment drives innovation by transforming advanced models into user-friendly simulation apps. He will also discuss leveraging rapid surrogate models for real-time insights and digital twin implementations, demonstrating how these interconnected tools democratize sophisticated engineering analysis and accelerate product development.

ABOUT THE INSTITUTE

Established in 1926, IIT(ISM) Dhanbad is located in India's coking coal belt, 260 km from Kolkata and 150 km from Ranchi. Spanning 393 acres, it offers world class facilities across 17 departments, providing courses in Engineering, Sciences, Management and Humanities. IIT (ISM) Dhanbad has played a vital role in the growth and development of India's mining, mineral and petroleum sectors. The institute holds a global rank of 20th and a national rank of 1st in Mineral and Mining Engineering (QS-2025) and an NIRF ranking of 15th in Engineering for the year 2024. IIT (ISM) Dhanbad debuted in the QS World University Rankings in the 701-710 band, notably ranking 1st in India and 21st globally in Mineral and Mining Engineering.

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JULY 01-05, 2026 | D2MMS | HYBRID

ABOUT THE PROGRAM

An international hybrid workshop and summer school exploring the intersection of physics-based simulation and data-driven methods, organized by the **Department of Mechanical Engineering**. This five-day event offers a structured platform to examine the role of data-driven approaches in mechanical engineering and allied fields, including mining, civil, and geotechnical engineering. Through expert-led lectures, tutorial sessions, and hands-on components, participants will engage with the mathematical, computational, and application-oriented foundations of modern modeling tools. As one of India's leading institutes in mining education and research, IIT (ISM) Dhanbad is proud to host this event, which will potentially contribute to the mining research ecosystem by introducing modern computational and data-driven tools for simulation, monitoring, prediction, and decision-making in mining and geotechnical applications.

Core Topics Include: Computational fluid dynamics (CFD), Computational mechanics, Digital twins, High-fidelity simulations, Reduced-order modeling, Robotics, Scientific machine learning, Potential applications of AI in mining.

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DISTINGUISHED SPEAKERS AND LECTURE TOPICS



Keynote Speaker

Dr. Christopher Rackauckas
*Applied mathematician, software engineer
Massachusetts Institute of Technology, USA*

Chris Rackauckas leads pioneering research in scientific machine learning (SciML), integrating differential equation modeling with artificial intelligence. He created Differential Equations.jl and is a founding developer of Pumas, enabling dramatic simulation speedups—including a 15,000x acceleration for NASA—and advancing personalized medicine as adopted by companies like Moderna. Chris is affiliated with MIT, Julia Computing, and Pumas-AI, and his work has earned him the Julia Community Prize and the Air Force AI Accelerator Excellence Award.



Prof. Romit Maulik
*Assistant Professor, Mechanical Engineering
Purdue University, USA*

Prof. Romit Maulik earned his PhD from Oklahoma State University (2019), then served as a Margaret Butler Postdoctoral Fellow and Assistant Computational Scientist at Argonne National Laboratory, and Assistant Professor at Penn State. His research focuses on multifidelity scientific machine learning for nonlinear dynamical systems in aerospace, Earth modeling, and fusion. His lecture covers spectral methods and forecasting chaotic dynamical systems with scientific machine learning.



Prof. Rajesh Ranjan
*Associate Professor, Aerospace Engineering
IIT Kanpur, India*

Prof. Rajesh Ranjan leads the Aeroscience Computations and Analysis Laboratory (ACAL). He earned his PhD from JNCASR under Profs. Narasimha and Deshpande, with prior degrees from IISc Bengaluru and BIT Sindri, and was a postdoc at Ohio State. His research spans CFD, GPU-accelerated solvers, reduced-order modeling, and physics-informed ML for turbulence and hypersonics. In his lecture on Scientific Machine Learning for Fluid Mechanics, he will cover fluid mechanics fundamentals, incompressible and compressible flows, CFD, and Physics-Informed Neural Networks (PINNs).



Prof. Suparno Bhattacharyya
*Assistant Professor, Mechanical Engineering
IIT (ISM) Dhanbad, India*

Prof. Bhattacharyya directs the D-cubed Lab, which advances modeling strategies for engineering systems at the nexus of computational mechanics, metamaterial design, and data-driven methods. He earned his PhD from Penn State University and completed postdoctoral research at Texas A&M University. His work develops intrusive and nonintrusive data-driven models for rapid, physically consistent simulations and digital twins. His lecture will explore how large-scale systems arise in engineering computation and why efficient representations matter for modern simulation and design.



Prof. Souvik Chakraborty
*Associate Professor, Applied Mechanics
IIT Delhi, India*

Prof. Souvik Chakraborty, focuses his research on developing scalable, interpretable, and trustworthy AI-driven methodologies at the intersection of scientific computing and machine learning. A Fulbright-Nehru Fellow and IIT Delhi Early Career Award recipient, his foundational work spans physics-informed machine learning, surrogate modeling, and stochastic dynamics. During his summer school session, his lecture will focus on AI for Solid Mechanics, where he will explore the concepts of Large-scale Systems, Physics Informed Neural Network, Neural Operator, and Agentic AI.



Prof. Giovanni Stabile
*Associate Professor, Numerical Analysis
Sant'Anna School of Advanced Studies, Italy*

Prof. Giovanni Stabile specializes in scientific machine learning, reduced-order modeling, and CFD. He holds a joint PhD from TU Braunschweig and the University of Florence, with previous posts at SISSA and Urbino. Creator of ITHACA-FV, miniDNN, and EZyRB, he leads the €1.45M ERC DANTE grant and collaborates with NVIDIA, Volkswagen, and Fincantieri. His lecture covers key methodologies including POD+NN, POD-RBF, Dynamic Mode Decomposition, and PINNs.



Prof. Aditi Sengupta
*Assistant Professor, Mechanical Engineering
IIT (ISM) Dhanbad, India*

Prof. Sengupta's research focuses on computational fluid dynamics, high-performance computing, fluid instabilities, turbomachinery aerodynamics, and the integration of machine learning with fluid mechanics. Her session examines generation of large-scale datasets in fluid dynamics and the emerging need for advanced data-driven models to transform raw simulation and experimental data into actionable physical insights, predictive understanding, and engineering knowledge.

Prof. Francesco Ballarin
*Professor, Mathematics and Physics
Università Cattolica del Sacro Cuore (Brescia & Milano), Italy*

Prof. Ballarin works in scientific computing, reduced-order modeling, CFD, and uncertainty quantification, with applications in fluid dynamics, biomechanics, and computational engineering. He earned his PhD from Politecnico di Milano in 2015 on reduced-order models for patient-specific haemodynamics and later contributed to the ERC-funded AROMA-CFD project at SISSA mathLab. His lecture will focus on reduced basis methods, computational fluid dynamics, and data-driven modeling for solids.



Prof. Omkar Mypati
*Assistant Professor, Mechanical Engineering
IIT (ISM) Dhanbad, India*

Prof. Mypati specializes in intelligent manufacturing, Industry 4.0, digital twins, robotics, and advanced manufacturing processes. His research focuses on AI-enabled systems, predictive maintenance, and integrating cyber-physical systems to enhance industrial productivity and reliability. During his session, he will demonstrate industrial robots performing automated tasks, highlighting real-time sensing, robot programming, and process monitoring in modern Industry 4.0 environments.